

Optimizing compensated-probe geometry below seven eighths

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A derivation relative to explicitly imported results

Abstract

We optimize two geometric parameters in the compensated-probe argument of the September 30, 2026 preprint *The Quasi-Riemann Hypothesis*. Assuming its specified generic arithmetic and analytic inputs and its finite-order Hecke seven-eighths theorem, we derive the strict zero-free half-plane

$$\operatorname{Re} s > \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799168740 \dots$$

for finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ and, by quadratic transfer, for all Dirichlet L -functions. We prove a variable- b low estimate with the complete additive Gram bound, give an exact continuous quadratic certificate for the central high estimate, and verify the detector capacities, Euler domains and outer rows. The critical certificate has zero additional endpoint margin; the common gap between the family supremum and the proposed boundary pays all real losses before the target character is chosen. The resulting parameters are optimal for the stated low intersection, count envelope and amplitude rectangle. A rational witness $40773/46600$ has strictly positive additional margin. The assertion allows the principal pole at 1 and excludes the boundary line. This is a cited-input derivation, not an independent certification of the imported proof or a new Lean certificate. No improvement of a critical-line zero proportion is asserted.

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1 Statement and exact imported hypotheses

Put $F = \mathbb{Q}(\sqrt{-3})$, and use the conventions of [1] for norm $q_a = N(a)$, ideal and element representatives, units, ray conditions and zero extension on nonunits. A superscript S deletes the finite Euler factors in the fixed arithmetic datum. Our previous paper [2] treats the boundary 69999/80000 with $b = 1/8$. We retain its imported package, but reprove every geometric condition affected by varying b . Neither paper certifies the imported package itself.

Definition 1.1 (Input package $\mathcal{R}$). The package consists of the following results of the pinned source [1], with their original coefficient classes, uniformities and side conditions: coefficientwise probe and Poisson identities, finite compensation, local arithmetic tables and ray calibration; smooth separation, Gaussian annuli, reflection sectors, generic reflected energy and the quantitative additive Gram bound; global and buffered reciprocal estimates, Hecke growth, functional equations and full contour tails; actual detector witnesses, marked and plain moments, recursive inverse moments, sixth-power amplification and uniform meshes; fixed-ray prime asymptotics; the whole finite-order Hecke seven-eighths zero-free theorem; and the quadratic factorization transferring the Hecke assertion to Dirichlet functions.

Define the whole-family supremum by

$$\beta_* = \sup(\{1/2\} \cup \{\operatorname{Re} \rho : L_F(\rho, \eta) = 0, \rho \text{ a nontrivial zero, } \eta \text{ primitive of finite order}\}).$$

The principal pole is not a zero in this definition. The seven-eighths input is used only as $\beta_* \leq 7/8$. In particular we do not assume the new half-plane is zero-free. We do not directly invoke a source low or shared-interface statement whose hypothesis fixes $b = 1/8$, $\ell = 1/6$ or a boundary at least $7/8$. We instead invoke the generic results and recheck their hypotheses. The source is frozen at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. All line numbers in Table 1 refer to its September 30 `build/paper.tex`.

Table 1: Imported inputs and the scope of their reuse.

Input	Source location	Conditions retained or rechecked
Smooth and Gaussian calculus	1123–1209, 1288–1315	One joint density, weighted seminorms and whole annular tails.
Reflection sectors and energy	2470–2506, 7663–7804	Actual residual dyads, bounded lengths, original masks and product coefficients; frozen powerful parts counted once.
Additive Gram	8340–8364	$Q, Y' \geq 1$, fixed polynomial ranges, $P_a = Y'^2/Q \geq 1$; complete third term retained.
Local and full-tuple identities	8707–9049	Selected primes operated on once; no division by general local corrections.
Absolute bounds and tails	9101–9195, 5672–5728	All heights and completed sums; fixed internal orders precede the external tail order.
Detector and moment inputs	4385–4547, 9220–9269, 12362–12389, 12492–12578	Actual witnesses, strict widths, finite coefficient group and zero-capacity cases.
Coefficients and capacities	15015–15446	Whole slots, conjugation, inducing exceptions, one shared conductor budget.
Whole-bin accounting	5808–5896, 6021–6155	Global move before local partitions; the same mean gain q .

Input	Source location	Conditions retained or rechecked
Principal and outer rows	6174–6301, 6342–6463	New analytic domain and $D_1(1/3)$ verified independently of the old numerical boundary.
Choices and bootstrap	16191–16463	Losses and mesh, then K , then principal μ , then target and height orders.
Height and continuation	6520–6580, 400–502, 6705–6780	Cutoff-independent probe, common saving, supremum contradiction and Dirichlet transfer.

Theorem 1.2. *Assume $\mathcal{R}$. Every finite-order Hecke L -function over F has no zero in $\operatorname{Re} s > \sigma_o$, where*

$$\sigma_o = \frac{1507 - 2\sqrt{921}}{1653}.$$

The same holds for every Dirichlet L -function, including ζ . Principal functions may have their pole at $s = 1$. The boundary line is not included.

The theorem is a relative implication under Definition 1.1. Its scope does not include independent verification of the source, its dependencies or its formal proof system. The simple critical-line proportion of [7] is not used in the bad-row count and is not improved here.

2 The same probe and the variable geometric scales

For fixed real parameters ℓ, b , write

$$l_x = \frac{1 - \ell - b}{2}, \quad l_y = \frac{1 - \ell + b}{2}, \quad M = 1 - \ell, \quad h = \frac{1 + 3\ell + b}{2}, \quad X = Z^{l_x}, \quad Y = Z^{l_y}. \quad (1)$$

Choose a fixed even K , equal positive $\ell_i = \ell/K$ and $P_i = Z^{\ell_i}$, with the original nonnegative nonzero smooth windows in disjoint subintervals of $(1, 2)$. Keep the source S, T, ξ , all permitted prime representatives, zero extensions, annular norm ratios and ray calibration. Write $I_\eta(X, Y, Z)$ for the source completed probe and $I_{\eta;D}$ for the same expression with $1_{D|cn^3}$ inserted. The marked operation in a slot is $\overline{\eta(p)} I_{\eta;p}(X, Y, Zq_p)$; the rescaled operation is $q_p^{-3/2} I_\eta(X/q_p, Y/q_p, Z)$, subtracted from it. If $p_J = \prod_{i \in J} p_i$ and $p_\emptyset = 1$, their compensated product is the independent physical expression

$$I_{\eta, \text{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \cdot I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}).$$

Each slot window remains at its own P_i even when another slot rescales a completed sum. Those completed sums are not finite; their absolute convergence is an input in $\mathcal{R}$. The compensation subsets and retained annular decompositions are finite.

The double residue at $w = 1, z = 1/6$ retains the exponent

$$C_b(s) = s + l_x/2 - 1 + h/6 = s - \frac{2}{3} - \frac{b}{6}. \quad (2)$$

The Gaussian factor at this residue is still $e^{(s-5/6)^2}$. There is no second normalizer introduced to improve the low bound.

3 The low estimate with free $\mathbf{b}$

Proposition 3.1. *Under the generic low inputs of $\mathcal{R}$, for fixed $1/6 \leq \ell \leq 1/5$ and $0 < b \leq 3(1 - \ell)/8$,*

$$|I_{\eta, \text{mod}}(Z)| \ll_{\eta, \text{fixed}, \varepsilon} Z^{(1-\ell)/4-b/6+\varepsilon}.$$

The arbitrary exponent loss can be prescribed before the target; constants and thresholds may depend on its fixed arithmetic data.

Proof. For a rescaled subset J put $d_J = \sum_{i \in J} \ell_i$, $\ell' = \ell - d_J$ and $M' = M - 2d_J$. Its tuple ratio $r_J = q_{p_J}/Z^{d_J}$ lies in a fixed positive compact interval, and the common row ball is

$$X' = XZ^{-d_J}/r_J, \quad Y' = YZ^{-d_J}/r_J, \quad Q = q_{b_*}X'Y' \asymp Z^{M'}, \quad M' + \ell' - 1 = -3d_J.$$

All surviving marked slots are summed inside B_m^J before taking the row norm. The common Gaussian profile and Fourier density are fixed before separating rows and tuples. Whole far annuli are first removed by the absolute completed-sum input and Gaussian decay.

Here is the geometric part of the reflected-energy calculation. Freeze the powerful part of length O and the actual squarefree dyad of length H . The source sectors give

$$\Delta_H = M' - O - H \geq -\varepsilon_Z, \quad 2A_0 \leq O + \varepsilon_Z, \quad N_0 \leq A_0, \quad z_a \leq \ell', \quad \varepsilon_Z = O_{\text{fixed}}(1/\log Z).$$

For retained Gaussian displacement θ_N the dual length is $T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0$. Subtracting $H - 3d_J$ gives

$$T_d - (H - 3d_J) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 - \theta_N \leq 2\varepsilon_Z + |\theta_N|.$$

For completeness the imported reflection expression, after counting frozen parts once, is

$$\begin{aligned} s_{\text{hyb}} &= \min\{v, z_a, (v + z_a)/3\}, \quad y_{\text{ref}} = v + 3\ell_b + e_\lambda \leq T_d + \tau_{\text{ref}}, \\ E_{\text{ref}} &= O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a \\ &\quad - s_{\text{hyb}} - \ell_b - 2e_\lambda/3 - (T_d - y_{\text{ref}})_+/2. \end{aligned}$$

Here v, ℓ_b are nonnegative dual lengths, S_0, B_0, N_0, A_0 are the frozen sector lengths, and a negative e_λ is only $O(1/\log Z)$. Retained support gives $\max(H, v + \ell_b) = H + O(\tau_{\text{ref}} + \kappa_G + \varepsilon_Z)$. In this formula, the branch $s_{\text{hyb}} = z_a$ gives exponent at most $M' - \Delta_H$. On the other branch $s_{\text{hyb}} \geq v/2$, so

$$s_{\text{hyb}} + \ell_b + 2e_\lambda/3 + (T_d - y_{\text{ref}})_+/2 \geq T_d/4 - O(\varepsilon_Z).$$

The exact frozen-part saving identity is

$$\begin{aligned} O/2 + \Delta_H + S_0 + B_0 + T_d/4 &= \frac{2M' + 2z_a - 1 - \ell' + 2\Delta_H - \theta_N}{4} \\ &\quad + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}. \end{aligned}$$

The second line is nonnegative. Using $z_a \leq \ell'$ and retaining the kernel saving before squaring and positive enlargement gives

$$M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H + \theta_N}{4} + \text{small losses}.$$

Since $1 + 3\ell' - 2M' = 5\ell - 1 + d_J$, both branches yield

$$\sum_{N m \ll Q} |B_m^J|^2 \ll Z^{M' + (5\ell - 1 + d_J)_+/4 + \varepsilon}. \quad (3)$$

This derivation does not contain b ; the original shared primes, empty surviving slots and powerful-part counting are unchanged.

The enlarged range still satisfies $M' \geq 1 - 3\ell > 0$ and $l_x - d_J, l_y - d_J > 0$. Thus $Q, Y' \geq 1$ eventually, in fixed polynomial ranges, and $P_a = Y'^2/Q = q_{b*}^{-1}Z^b \geq 1$ eventually. The generic Gram bound, including its third term, is

$$\sum_{N m \ll Q} |A_m(Y')|^2 \ll (Q/Y')(1 + P_a^{1/6} + P_a^2/Y')Z^\varepsilon$$

with the original ray/Gauss coefficients and fixed height seminorm cost. Cauchy, including the separation factor $Q^{-1/2}$, bounds a fixed tuple by

$$(X')^{1/2} Z^{\frac{1}{2} \max\{0, b/6, 2b - l_y + d_J\} + (5\ell - 1 + d_J)_+ / 8 + \varepsilon}.$$

Tuple number, the original $q_{p,J}^{-3/2}$ coefficient and the shortening of $(X')^{1/2}$ contribute respectively $d_J, -3d_J/2$ and $-d_J/2$. Relative to $l_x/2 + b/12$, their combined additional exponent is

$$F(d) = -d + \frac{1}{2}(11b/6 - l_y + d)_+ + \frac{1}{8}(5\ell - 1 + d)_+.$$

Every piece has slope at most $-3/8$, so its maximum is at $d = 0$. Because $\ell \leq 1/5$, the resulting full low exponent is

$$L_{\text{low}} = \frac{l_x}{2} + \frac{b}{12} + \frac{1}{2}(11b/6 - l_y)_+ = \max \left\{ \frac{1 - \ell}{4} - \frac{b}{6}, \frac{b}{2} \right\}. \quad (4)$$

The stated range of b selects the first branch. Small reflection and separation losses, finitely many subsets, summable joint Fourier seminorms and logarithmic dyad counts can be absorbed into the prescribed ε in their original order. $\square$

For our parameters the stronger all-subset condition

$$l_y - \ell - \frac{11b}{6} = \frac{1 - 3\ell}{2} - \frac{4b}{3} > 0$$

also holds. We have proved the wider low range, rather than assuming this stronger condition is necessary. When $b \leq 0$, the present direct use of the Gram input $P_a \geq 1$ is not justified; no such extension is used. On the first branch of (4), the residue intersection is

$$C_b(\sigma) = \frac{1 - \ell}{4} - \frac{b}{6} \iff \sigma = \frac{11}{12} - \frac{\ell}{4}. \quad (5)$$

Outside that branch one cannot keep this same intersection.

4 Actual counts and the continuous endpoint certificate

Use the original legal $\kappa = 3/4$, justified by $\beta_* \leq 7/8$, and the algebraic envelope $\alpha = 5/6$. For nonfloor bins set $\delta = 2a - 1 \in [1/50, 3/4]$. Let q be the length-weighted mean amplitude over all slots; error and zero slots have gain zero. Put

$$x = q/\delta \in [0, 1/2], \quad y = 1/2 - x, \quad D = \frac{37 + 34y}{18}, \quad P = \frac{7 + 18y + 8y^2}{9},$$

$$J = (5/6 - \delta)D + \delta P, \quad R_* = 1 - \delta + \frac{(5/6 - \delta)\delta P}{2J}. \quad (6)$$

The actual inverse/plain crossing gives these functions with $35/54 \leq J \leq 5/2$ and $1 - \delta \leq R_* \leq 1 - 2\delta/3$. Section 6 verifies that the new geometric scales meet its arithmetic prerequisites.

At (5), the central endpoint accounting is

$$E_\sigma(h) = -\frac{1}{4} + \frac{5\ell}{4} + \frac{b}{6} + (\frac{1}{2} + \ell)\delta + \ell\delta x - h(1 - R_*). \quad (7)$$

In particular $\partial_b E_\sigma = R_*/2 - 1/3$. Clearing denominators gives $-1296JE_\sigma = A(y)\delta^2 - B(y)\delta + C(y)$, where

$$\begin{aligned} A(y) &= 252 + 756\ell - 576b + (648 + 288\ell + 720b)y \\ &\quad + (288 + 1008\ell + 864b)y^2 + 1152\ell y^3, \\ B(y) &= 624 - 1440\ell - 1176b + (504 - 420\ell - 456b)y + (-48 + 120\ell + 432b)y^2, \\ C(y) &= 555 - 2775\ell - 370b + (510 - 2550\ell - 340b)y. \end{aligned} \quad (8)$$

For $Q = 4AC - B^2$ its constant coefficient is

$$Q(0) = -530496b^2 + 1887840b\ell - 184032b - 10465200\ell^2 + 678240\ell + 170064.$$

For fixed ℓ , its maximum in b occurs at $b_{\text{opt}} = (2185\ell - 213)/1228$ and equals $-1631700(1653\ell^2 - 66\ell - 35)/307$.

Proposition 4.1. *Set*

$$\ell_\circ = \frac{33 + 8\sqrt{921}}{1653}, \quad b_\circ = \frac{2185\ell_\circ - 213}{1228} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}.$$

Then $E_{\sigma_\circ}(h) \leq 0$ on the entire rectangle $\delta \in [1/50, 3/4]$, $y \in [0, 1/2]$. Its unique equality point is $y = 0$, $\delta_\circ = (49 - \sqrt{921})/48$.

Proof. Write $e = \ell_\circ$. It satisfies $1653e^2 - 66e - 35 = 0$ and $1/6 < e < 167/1000$, $123/1000 < b_\circ < 124/1000$. The coefficients of A are

$$\frac{108036 - 82548e}{307}, \quad \frac{160596 + 481716e}{307}, \quad \frac{42408 + 781416e}{307}, \quad 1152e,$$

all strictly positive. Reducing Q by the same quadratic identity gives

$$\begin{aligned} Q_0 &= 0, \\ Q_1 &= 17404800(5465 - 31728e)/169157, \\ Q_2 &= 1200(47597384 - 214807491e)/169157, \\ Q_3 &= 4800(131691060e - 18009311)/169157, \\ Q_4 &= 14400(1377691e - 209998)/8903. \end{aligned}$$

The isolating bounds above imply $Q_i > 0$ for $i = 1, \dots, 4$. Thus, for all $y \geq 0$,

$$A\delta^2 - B\delta + C = A\left(\delta - \frac{B}{2A}\right)^2 + \frac{Q}{4A} \geq 0.$$

Since $J > 0$ on the actual rectangle, this proves the assertion. Equality forces $y = 0$ and $\delta = B(0)/(2A(0)) = (49 - \sqrt{921})/48$, which lies strictly between $1/50$ and $3/4$. This is a continuous coefficient-positivity proof, not a grid inference. Equation (5) gives (2) at the boundary stated in Theorem 1.2. $\square$

Proposition 4.2 (The precise parameter optimality). *No $\ell > \ell_\circ$ and any b can make all endpoints nonpositive while retaining (5), the count (6) and the complete stated amplitude rectangle.*

Proof. At the fixed point $y = 0, \delta = \delta_\circ$, the count gives $R_* = 2/3$. Thus (7) is independent of b and equals

$$-\frac{5}{12} + \frac{3\ell}{4} + (\frac{1}{2} + \frac{3}{2}\ell)\delta_\circ.$$

It vanishes at $\ell_\circ$ and its slope in ℓ is $3/4 + 3\delta_\circ/2 > 0$. $\square$

This is optimality of a particular low/count certificate, not of all possible probes or L -function theory. We do not claim that actual bad rows attain the envelope or that a corresponding zero exists.

5 A rational witness and comparison with fixed b

For $\ell_r = 1/6 + 1/5825 = 5831/34950$ and $b_r = 617/5000$,

$$l_{x,r} = \frac{2480617}{6990000}, \quad l_{y,r} = \frac{3343183}{6990000}, \quad h_r = \frac{1891861}{2330000}, \quad \sigma_r = \frac{40773}{46600}.$$

At these parameters A, C, Q have strictly positive coefficients. Moreover $A_0Q(y) - Q_0A(y)$ has nonnegative coefficients, and hence

$$-E_{\sigma_r}(h_r) \geq \frac{Q_0}{12960A_0} = \frac{10580567}{347281513800000} > 10^{-8}.$$

Here completing the square first gives $Q/(4A)$, and $J \leq 5/2$ supplies the factor 12960. This does not mistake $A(0)$ for an upper bound on $A(y)$. The exact coefficients are in the reproducible algebra record [6].

The fixed- b critical value in [3] has $\sigma_c = 7/8 - t_c/4$, where $1/5842 < t_c < 1/5841$ is the positive root of

$$28224 - 164747520t - 669772800t^2 = 0.$$

Since $1/5825 > 1/5841$, $\sigma_r < \sigma_c$. Also $1653\ell_r^2 - 66\ell_r - 35 = -9289/407167500 < 0$; the quadratic is increasing on $[1/6, \infty)$, so $\ell_r < \ell_\circ$ and $\sigma_\circ < \sigma_r$. Both comparisons are exact. For orientation only,

$$\sigma_\circ = 0.874957069799168740\dots, \quad \sigma_c = 0.874957200615373761\dots$$

We prove the theorem at the critical algebraic value, not only at the rational witness.

6 Actual arithmetic admission and all row ranges

For a physical row length d the same-source accounting reads

$$E_\sigma(d) = K_\sigma + (\frac{1}{2} + \ell)\delta + \ell q - h(1 - R) + (d - h)(R + \delta/2 - 17/50), \quad K_\sigma = 2/3 + \ell + b/6 - \sigma. \quad (9)$$

Expanding the original unfactored exponent $a - \sigma + h(17/50 - 1/6) - al_y - \ell/2 + \ell q + d(R + \delta/2 - 17/50)$ gives this formula. The joint bound for strict local errors and the numerator uses one conductor deficit once; its gain remains ℓq . There is no second charge for each ramified error label.

For nonfloor selected rows $1/2 \leq d \leq h$, the slope is at least $33/50 - \delta/2 \geq 57/200 > 0$, so the endpoint controls the whole range. The floor $a = 51/100$ uses $\delta = 1/50$, $R = 1$, $q \leq 1/100$ independently of a witness. The no-slot count on $1/100 \leq d \leq 1/2$ uses $R = 76/75 - 2\delta/3$,

covering the floor, with positive slope $101/150 - \delta/6$. At $\ell = e = \ell_o, b = b_o$, direct calculation gives

$$\begin{aligned} l_y - e - 11b/6 &= (1347 - 7133e)/1842 > 2/25, \\ 5e - h &= (6411e - 1015)/2456 > 1/50, \\ E_{\text{floor}}(h) &= (290401e - 49533)/184200 < -1/200, \\ E_{\text{middle}} &= (292151e - 84703)/1473600 < -1/50. \end{aligned} \tag{10}$$

The middle bound follows from $q \leq \delta/2$: its value at $d = 1/2$ is $-73/300 + (13/50)\ell - (49/300)b + \delta(1/6 + 3\ell/4 - b/4)$, maximized at $\delta = 3/4$.

Assume for contradiction $\beta_* > \sigma_o$, and put $\Delta = \beta_* - \sigma_o > 0$. The bootstrap gives $\Delta \leq (e - 1/6)/4$. Choose $\zeta = \Delta/32$. Then $\zeta < 1/384000$ and (10) implies

$$\ell/(h + \zeta) > 1/5 > 7/37, \quad h + \zeta < 1.$$

The selected-row slope is at most 2, so the extension from h to $h + \zeta$ costs at most $\Delta/16$. The actual comparison is with $C_b(\beta_*)$:

$$E_{\beta_*}(d) = E_{\sigma_o}(d) - \Delta. \tag{11}$$

Proposition 4.1 thus provides central saving at least $15\Delta/16$ before small real losses. Zero additional margin at the critical endpoint does not preclude continuation.

We spell out the moment prerequisites that varying b must preserve. The adaptive parameter is $t = 1 + \delta P/(2J)$. The original witnesses have $r + m \geq t - o(1)$, $r \leq t + o(1)$, $m \leq 1/2 + o(1)$ and the original squared spike sizes. Writing $x = q/\delta$, the original inverse/plain threshold is

$$r_*(t) = \frac{(2 - 8x/9)t - 5x/9}{3 - 17x/9}, \quad r_*(t) \geq 23/37, \quad 1 \leq t \leq 3/2.$$

On the inverse branch $r \geq r_*(t)$, $r < 1$ and positive $z_M = (1 - r)/2$. For a fixed capacity decrement $\nu_0 > 0$, requesting $z \leq z_M - \nu_0$ gives the two inverse widths

$$1 - r - 2z \geq 2\nu_0, \quad 3 - 2r - 8z \geq 8\nu_0 + 9/37 - o(1).$$

On the plain branch $r \leq r_*(t)$, $m < 1/2$ and positive $z_P = (1 - 2m)/(6\kappa)$; requesting $z \leq z_P - \nu_0$ gives, with $m \geq 1/3 - o(1)$, $1 - 2m - 6\kappa z \geq 6\kappa\nu_0$. Maximum demands are $z_M \leq 7/37$ and $z_P \leq 2/27 + o(1)$. The supply above therefore suffices; it does not itself strengthen the moment theorem. Long-witness and zero-capacity endpoints retain the original zero-slot or sixth-power cases.

Only $d \geq 1/2$ uses selected primes, so their physical lengths $w_i = \ell_i/d \leq 2\ell_i$. Choose losses, decrement, mesh and rounding before K ; then every w_i can be smaller than the fixed plain mesh. The greedy selection loses at most one whole-slot length and uses the gain averaged over all slots. Coefficients remain the fixed finite Θ combination of the original arithmetic phases; all witnesses and selected factors are conjugated together on the plain side. The physical slots remain disjoint from the internal plain pool. Outside- S ramification excludes the original inducing exceptions; the remaining S -supported sixth-power-free rows are finite and belong to the independently handled bounded rows. These are precisely the prerequisites of the unchanged count (6).

7 Euler domains, principal residue and outer rows

Varying b does not alter a local coefficient or zero extension. The full selected-prime operation and the coefficientwise physical identity remain the source identities. The resulting raw Mellin weight is

$$X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z) \widehat{W}_1(w).$$

The full tuple is the finite selected-prime sum with selected factors G_p and the normally convergent unselected correction product. It is never defined by dividing by a general H_p .

The same local finite-table calculation as in [2] applies on the enlarged principal box:

$$\operatorname{Re} s \geq \sigma_o, \quad \operatorname{Re} w \geq 19/20, \quad \operatorname{Re} z \geq 33/200.$$

Indeed $\sigma_o > 5/6 > 401/600$. The good and ramified defects are

$$\begin{aligned} c_{\text{good}} &= \min\{6\sigma_o - 401/100, \sigma_o - 3/50, 47/50\} = \sigma_o - 3/50 > 81/100, \\ c_{\text{ram}} &= \min\{\sigma_o - 1/20, 3\sigma_o - 3/2\} = \sigma_o - 1/20 > 82/100. \end{aligned}$$

The denominators $1 - R, 1 - V, 1 - D$ stay away from zero. The general good local term with exponent $4 - 6s - 6z$ retains the correction $\theta = (-\operatorname{Re} w)_+$. In $D_1(\varepsilon_0)$ the exponent used for the correction is $\varepsilon_H = \min(\varepsilon_0, 1/50)$, not always ε_0 . These strict margins imply local normal convergence and full height majorants, not just values on a single contour.

Whole-bin contours are moved on the global line before amplitude and error partitions, followed by the original buffered local move. All strict labels use the joint numerator estimate once. The principal correction is the original normally convergent product

$$H_\eta(s) = \prod_{p \notin S} H_p(s, 1, 1/6) = \mathcal{H}_{\eta,1}(s, 1, 1/6),$$

with the same final S as the physical expression. For the principal row choose a target-independent P_0 such that the positive product majorant gives $|H_\eta(s) - 1| \leq 1/2$ on $\operatorname{Re} s \geq \sigma_o$. Enlarging the final S only shrinks this majorant. More explicitly, put

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6} \sim c_i P_i^{1/6} / \log P_i, \quad A_T = (-1)^K Z^{-\ell/6} \prod_i S_i(Z).$$

The fixed-ray calibration supplies $c_i > 0$, the same $c_S = \widehat{W}_1(1)M(1/6)(\operatorname{Res}_{v=1} \zeta_F^S(v))^2/6 > 0$, and eventually nonzero $|A_T| \asymp (\log Z)^{-K}$, with subpower inverse. These are the same finite S and actual ray slots used in the physical expression.

The principal contours start with global s , then pass the residues at $w = 1, z = 1/6$, with final nonresidue lines $19/20$ and $33/200$. The double-residue s line is moved only to the right, to 2. The three error margins are

$$m_w = l_y/20 > 0, \quad m_z = h/600 > 0, \quad 0 < \mu < \sigma_o \min_i \ell_i.$$

The four principal local error exponents remain $-\operatorname{Re} s, -6\operatorname{Re} z, 4 - 5\operatorname{Re} s - 6\operatorname{Re} z, 1 - \operatorname{Re} w - 6\operatorname{Re} z$; their weakest decay at the residue is σ_o .

Small rows $d < d_{\min} = 1/100$ and bounded nontrivial units are handled separately. The old numerical invocation of $D_1(3/8)$ is replaced by the local proof on $D_1(1/3)$, valid since $\sigma_o + 1/2 > 4/3$. The full absolute tuple and numerator bound give, relative already to $C_b(\beta_*)$,

$$E_{\text{small}} = h(13/75) - l_y/2 + 63/5000 = \frac{2020475e - 1127329}{9210000} < -2/25.$$

No additional Δ is charged to this formula. Large rows $U > Z^{h+\zeta}$ use the original full absolute sum on $(2, 2, z_\infty)$:

$$Z^{B_0+(h+\zeta)(1+\varepsilon)-\zeta z_\infty+\varepsilon}, \quad B_0 = l_x/2 + 1 + l_y = 7/4 - 3\ell/4 + b/4.$$

After fixing ζ , choose a fixed sufficiently large z_∞ before the target; then small losses and all internal height orders. This gives any prescribed saving without changing an order with a moving row or with Z .

8 Uniform choices and whole-family continuation

Fix the critical geometry and the common contradiction gap Δ before any target character. Use small fixed fractions of Δ for moment losses, strict capacity decrement, uniform mesh and rounding. Then choose even K and the disjoint windows. Only afterwards put $\mu = \sigma_\circ \ell / (2K) > 0$ and choose amplitude widths, prime losses and the small contour displacement. Principal losses are separately smaller than $m_w/4, m_z/4, \mu/4$; central total real losses are below $\Delta/4$, and floor, middle and small losses leave half their respective independent margins. There is no circular selection of K using a μ that depends on K . Fix ζ and z_∞ in the same target-independent phase.

Let

$$m_0 = \min\{\Delta, m_w, m_z, \mu, 1/200, 1/50, 2/25\} > 0, \quad m = m_0/4.$$

The common low loss, including the inverse normalizer, is $\omega = \Delta/2 < \Delta$. For a chosen target, fix its arithmetic datum and final S , and use that same datum for every physical expression, calibration and correction. For sufficiently large Z define the cutoff-independent normalized physical function J_η ; for all $Z > 0$ define f_η :

$$J_\eta(Z) = I_{\eta, \text{mod}}(Z)/(c_S A_T(Z)), \quad f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C_b(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds.$$

The low bound gives $|J_\eta| \ll_\eta Z^{C_b(\sigma_\circ) + \omega}$. The principal and all high estimates, with the same S and normalizer, give finite A_η, B_η such that

$$|J_\eta - f_\eta| \ll_{\eta, N} Z^{C_b(\beta_*) - m} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}. \quad (12)$$

Neither function contains T_1 . The internal moment, Sobolev and seminorm orders giving A_η, B_η are fixed before the external N . All retained frequencies share one cumulative $T_1/2$ allocation.

For the actual detector ceiling $\tau_{0, \eta} > 0$ choose

$$0 < \tau_\eta \leq \min\{d_{\min}/100, \tau_{0, \eta}, m/[4(A_\eta + 1)]\}, \quad T_1 = Z^{\tau_\eta}.$$

After this choose N_η so that $B_\eta - N_\eta \tau_\eta < C_b(\beta_*) - m/2$, and only then the threshold in Z . Equation (12) has a common target-independent saving $\sigma_{\text{hi}} = m_0/8 > 0$. The target dependence of τ_η, N_η , constants and thresholds does not change this saving.

Proof of Theorem 1.2. All prerequisites for the source continuation argument have now been re-established at the new geometry. For clarity we describe its contradiction step. The Gaussian and rightward shifts give rapid decay of f_η as $Z \rightarrow 0$. No small- Z definition of J_η or inverse A_T is needed. The common high saving and the strict low gap imply locally uniform convergence of the Mellin/Fourier identification to a line left of β_* , by the common amount $\varepsilon_* = \min\{\Delta/2, m_0/8\} > 0$. Specifically, $\mathcal{F}_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C_b(s)} dZ/Z$ converges locally uniformly on $\text{Re } s > \beta_* - \varepsilon_*$. On the initial line 2 the Gaussian Fourier identification gives $\mathcal{F}_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s, \eta)$. Analytic continuation, without moving a contour across a target zero, therefore makes $H_\eta/L_F^S(\cdot, \eta)$ holomorphic to that line. The positive product choice $|H_\eta - 1| \leq 1/2$ excludes cancellation by the numerator.

The supremum β_* need not be attained: its definition gives a target zero with real part greater than $\beta_* - \varepsilon_*$. All the preceding choices are common before that target, and the resulting holomorphic reciprocal contradicts this zero. Thus $\beta_* \leq \sigma_\circ$. Finite Euler deletions have no zero for $\text{Re } s > 0$. The quadratic Dirichlet factorization in $\mathcal{R}$ transfers the assertion to all Dirichlet functions. At $s = 1$ retain the principal pole and use $L(1, \chi_{-3}) = \pi/(3\sqrt{3}) \neq 0$ for the zeta factor. $\square$

9 Relation to zero proportions and further work

The argument does not use the simple critical-line proportion of Alpöge–Furman [7]. Its count improvement is geometric within a different actual family and moment calculation. Conversely, a narrower global zero-free strip alone does not pay the signed four-point correlation in the original finite critical-line response. Our companion note [4] keeps the natural masks, ghost poles and finite frame while proving a canonical Type-I admission and low-prime fourth-norm payment. Its conditional one-factor contour estimate is weaker than a local unconditional finite-ratio Gram bound; neither supplies a new fourth-moment constant or a larger simple-zero proportion.

Proposition 4.2 identifies the next boundary task: one must strengthen the actual count/gain relation or the admissible moment mechanism near $y = 0, \delta = \delta_o$, rather than only changing b . For proportions the remaining task is the original joint signed response, including the common centering, cross terms, all relevant cells and finite-frequency alias bookkeeping. Neither unfinished task is replaced by the optimality statement.

A Exact algebra and verification scope

The pinned source canonical LF SHA-256 is

42a5ee0febca59fd1def55cfd6c6808c322ef4237d7726303ea52f711deac6a3.

The derivation is recorded in note 449 [5]. The independent parameter report and exact script [6] use rational arithmetic and $\mathbb{Q}(\sqrt{921})$, with no floating-point inference. The script checks the displayed discriminant coefficients, positive signs, equality point, rational witness and geometric margins. Its 841 checks include 189 direct endpoint cases as implementation checks. The continuous proof is the polynomial identity and positivity argument in Section 4, not those cases.

Run, from the repository root,

```
python scripts/hybrid_free_b_geometry_exact_audit.py
```

The script prints its own canonical hash and an exact JSON record. Final source, PDF and independent full-paper review hashes are bound separately in the build manifest. The script does not certify actual prime sums, imported estimates, infinite contours, the source proof kernel or RH. The earlier 69999/80000 paper is preserved as its delivered version; the present result is a separate paper.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis*, September 30, 2026 preprint, pinned source at [OpenAI/math](#), commit [adc7f1241](#).
- [2] RH–Weil research project, *A small improvement of the seven-eighths zero-free boundary from a compensated probe*, October 7, 2026. [papers/seven-eighths-boundary-improvement-paper.tex](#).
- [3] RH–Weil research project, Note 447, October 7, 2026. [notes/447-larger-slot-discriminant-and-critical-strip.md](#).
- [4] RH–Weil research project, Note 448, October 7, 2026. [notes/448-canonical-type-i-admission-finite-gram-and-low-prime-fourth-norm.md](#).

- [5] RH–Weil research project, Note 449, October 7, 2026. `notes/449-free-b-compensated-geometry-and-optimal-relative-boundary.md`.
- [6] RH–Weil research project, Free- b optimization and exact audit, October 7, 2026. `reviews/2026-10-07/hybrid-free-b-geometry-optimization.md`; `scripts/hybrid_free_b_geometry_exact_audit.py`.
- [7] L. Alpöge and R. Furman, *More than two thirds of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2608.13637v2, 2026, [version 2](#).